WORKED SOLUTION

STEP I 1994, Question 10

Mechanics • Circular Motion

> **STEP I 1994, Question 10**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/94-s1-q10

The key idea

This is a conical pendulum with a stretchy string. The particle has no vertical acceleration, so the vertical component of the tension balances the weight, while the horizontal component supplies the centripetal force. The new feature is that the radius of the circle depends on the extension, which depends on the tension, so Hooke's law ties everything together. The inequality then comes from asking which values of $\cos\theta$ actually describe a circle.

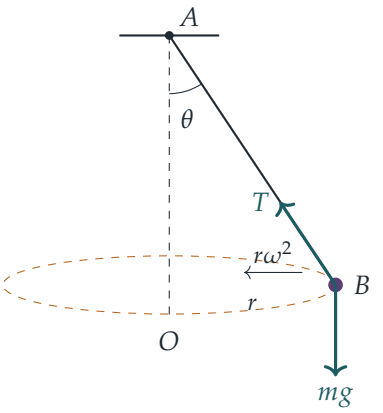
Solution

Finding $\cos\theta$

Model B as a particle and AB as a light elastic string obeying Hooke's law. Let the extension be x and the tension T, so

$$T = \frac{\lambda x}{l}, \qquad AB = l + x.$$

The particle moves in a horizontal circle of radius $r = (l + x)\sin\theta$ about the point O vertically below A. The only forces on it are its weight and the tension along BA.



Resolving vertically (no vertical acceleration):

$$T\cos\theta = mg. \tag{1}$$

Resolving horizontally, towards the centre O (acceleration $r\omega^2$):

$$T\sin\theta = m\omega^2(l+x)\sin\theta.$$

For a genuine circle $\sin\theta \neq 0$, so we can divide by it:

$$T = m\omega^2(l+x). \tag{2}$$

Now Hooke's law gives $\lambda x/l = m\omega^2(l+x)$, that is,

$$x\left(\lambda - m\omega^2 l\right) = m\omega^2 l^2.$$

Since $x > 0$ (from (1), $T > 0$, so the string is stretched) and $m\omega^2 l^2 > 0$, we have $\lambda - m\omega^2 l > 0$ and

$$x = \frac{m\omega^2 l^2}{\lambda - m\omega^2 l}, \qquad l + x = \frac{l\lambda}{\lambda - m\omega^2 l}.$$

Then (2) gives

$$T = \frac{m\omega^2 l\lambda}{\lambda - m\omega^2 l},$$

and (1) gives

$$\boxed{\cos\theta = \frac{mg}{T} = \frac{g\left(\lambda - m\omega^2 l\right)}{l\lambda\,\omega^2}}$$

When the motion is possible

Ask yourself What does each restriction on $\cos\theta$ say physically?

For the particle to move in a horizontal circle below A, with the string taut, we need $0 < \theta < \frac{1}{2}\pi$, i.e. $0 < \cos\theta < 1$. Conversely, if ω gives $0 < \cos\theta < 1$ from the formula,

then working back through the steps (with $\lambda - m\omega^2 l > 0$) produces a positive extension and tension satisfying (1) and (2), so the motion does exist.

Upper bound. $\cos\theta > 0$ holds exactly when $\lambda - m\omega^2 l > 0$, since the other factors are positive. This is

$$\omega^2 < \frac{\lambda}{ml}.$$

Physically: if ω is too large, no finite extension of the string can supply the required centripetal force, since the extra length needs extra force faster than the string provides it.

Lower bound. $\cos\theta < 1$ means

$$g\lambda - gm\omega^2 l < l\lambda\omega^2 \iff g\lambda < \omega^2 l(\lambda + mg) \iff \omega^2 > \frac{g\lambda}{l(\lambda + mg)},$$

where we divided by $l\lambda\omega^2 > 0$ first and by $l(\lambda + mg) > 0$ last, so the inequality directions are kept. Physically: if ω is too small, the formula would demand $\cos\theta \geq 1$; the only option is for the particle to hang at rest on the axis ($\theta = 0$), which is not circular motion.

Putting the two together, the motion is possible only if

$$\frac{g\lambda}{l(\lambda + mg)} < \omega^2 < \frac{\lambda}{ml},$$

as required.

Checks

- At the lower bound, $\cos\theta = 1$: the string is then vertical with tension mg and length $l(\lambda + mg)/\lambda$, which is exactly the equilibrium length of the hanging particle. Note $\omega^2 = g/(\text{that length})$, the familiar conical-pendulum threshold $\omega^2 > g/L$.
- As $\lambda \to \infty$ (inextensible string) the result becomes $\cos\theta = g/(l\omega^2)$, the standard conical pendulum, and the range becomes $\omega^2 > g/l$.
- Numerically, with $g = 9.8$ and random m, l, λ and ω, solving Hooke's law and the horizontal equation directly by bisection gives the same $\cos\theta$ as the formula, and both resolved equations balance; across 20 000 random cases the motion exists exactly when the inequality holds.

Common slips

- Using radius $l\sin\theta$ instead of $(l + x)\sin\theta$: the string is stretched.
- Forgetting to justify dividing by $\sin\theta$, and so missing why $\theta = 0$ (and hence $\cos\theta = 1$) is excluded.
- Only proving $\cos\theta < 1$ and forgetting that $\cos\theta > 0$ (equivalently, a positive extension) gives the upper bound.

Beyond STEP

The upper bound is a simple example of a *runaway* or instability: as $\omega^2 \to \lambda/(ml)$ the required extension blows up. The same effect appears for rotating elastic rods and in the stability of spinning bodies, and the lower bound is a bifurcation, where the hanging equilibrium gives way to circular motion. For more on rotating-frame mechanics, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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